

Pre-class preparation

Please watch the following video:

- Video: <https://expl.ai/GYKDYQF>

Objectives

By the end of the day's class, students should be able to do the following:

- State the definition of the Null and Alternative hypotheses in the language of parameter spaces, both for simple and composite hypotheses, and in terms of one- and two-sided hypotheses.
- Give the definition of the test procedure, the critical region and the test statistic for a hypothesis test.
- Conduct hypothesis tests in a variety of frameworks for real data.
- State the definition of the power function and explain how it relates to the critical region and the probability of Type I and Type II errors.

Reflection Questions

Please submit your answers to the following questions to the corresponding Canvas assignment by 10:00AM:

1. Suppose $X_1, \dots, X_n | \mu \stackrel{\text{iid}}{\sim} N(\mu, 1)$ with $\mu \in \mathbb{R}$ unknown, and that we wish to test the following hypotheses:

$$H_0 : \mu = 0 \quad H_1 : \mu \neq 0$$

Consider the following test procedures:

- δ_1 : Reject H_0 if $\bar{X} \geq 2$
- δ_2 : Reject H_0 if $|\bar{X}| \geq 2$

- (a) In δ_1 and δ_2 above, are we defining critical regions or rejection regions?
 - (b) Which testing procedure δ_1 or δ_2 would you prefer? Briefly explain why.
2. In your own words, summarize the relationship between the partition of the parameter space into subsets Ω_0 and Ω_1 and the partition of sample space into subsets S_0 and S_1 .
 3. (Optional) Is there anything from the pre-class preparation that you have questions about? What topics would you like some more clarification on?